Gabriele Benedetti

May 22, 2026 | https://github.com/gbene |
https://gabri.xyz | mail@gabri.xyz

Education

2025 - present	PhD in Geophysics *University of Iceland* — *Reykavik, Iceland*	
2020 - 2022	**MSc in Geology and Geodynamis** *University of Milano-Bicocca* — *Milano, Italy*	(110/110 cum laude)
Thesis	New tools for Digital Outcrop Models analysis: Implementation for the PZero software	
Keywords	Structural geology data analysis, Full stack development, Principal Component Analysis, Automatic segmentation, Clustering algorithm, VTK	
2017-2020	**BSc in Geological and Geotechnical sciences** *University of Milano-Bicocca* — *Milano, Italy*	(107/110)
Thesis	Photogrammetric techniques applied to invertebrate paleontology	
Keywords	Photogrammetry, Agisoft Metashape, Preservation and archival	

Work Experience

University research assistant Feb 2023 - Aug 2025

University of Milano-Bicocca — Milano, Italy

- Point cloud segmentation procedures for fracture planes extraction
- Right censoring bias estimation and correction for fracture length statistical modelling
- Stochastic DFN parameter calibration

Programmer May 2022 - Feb 2023

Pro Iter Ambiente s.r.l. — Milano, Italy
Development of the PZero software

- User interface and quality of life improvements
- Backend refactorization
- Wells and geophyisical profiles integration
- Streamline input and output with CAD/BIM environments

Programmer Nov 2019 - Dic 2019

Freelance — Milano, Italy

Designed and programmed an Agisoft Metashape compatilbe pipeline to calculate the elevation difference between two models.

Publications

[1] Gabriele Benedetti and Elías Rafn Heimisson. "HighSeas.jl Accelerating rate and state simulations with GPUs". In: *Computational Geosciences* (Submitted).

[2] Stefano Casiraghi, Daniela Bertacchi, Gabriele Benedetti, Silvia Mittempergher, Federico Agliardi, Mattia Martinelli, Francesco Bigoni, Cristian Albertini, and Andrea Bistacchi. "Statistical Assessment of the Representative Elementary Area for Areal Fracture Intensity (P21) in Digital Outcrop Models". In: *EGUsphere (preprint)* (2026), pp. 1–40. URL: https://egusphere.copernicus.org/preprints/2026/egusphere-2026-2409/.

[3] Gabriele Benedetti, Stefano Casiraghi, Daniela Bertacchi, and Andrea Bistacchi. "Unbiased statistical length analysis of linear features: adapting survival analysis to geological applications". In: *Solid Earth* 16.4/5 (2025), pp. 367–390.

[4] Stefano Casiraghi, Gabriele Benedetti, Daniela Bertacchi, Silvia Mittempergher, Federico Agliardi, Bruno Monopoli, Fabio La Valle, Mattia Martinelli, Francesco Bigoni, Cristian Albertini, et al. "An integrated workflow for parametrization of fracture network geometry in digital outcrop models". In: *Solid Earth* 16.11 (2025), pp. 1351–1382.

[5] Ivan Kosović, Bojan Matoš, Stefano Casiraghi, Gabriele Benedetti, Tihomir Frangen, Kosta Urumović, Ivica Pavičić, Andrea Bistacchi, Silvia Mittempergher, Marco Pola, et al. "Hydrogeological parameterisation of the Daruvar thermal aquifer: integration of fracture network analysis and well testing". In: *Geologia Croatica* 77.2 (2024), pp. 99–125.

Awards and Grants

2026 **Erasmus+ traineeship** (ML4SEG)

Machine learning for Solid Earth Geosciences and Earthquake Physics

2024 **Best student presentation award** (DRT24 — *Barcelona, Spain*)

Scale independent unbiased statistical length analysis of linear features: Adapting survival analysis techniques to geological applications

2024 **Outstanding Student Presentation and Poster award** (EGU24 — *Wien, Austria*)

FracAbility: A python toolbox for survival analysis in fractured rock systems

Conferences

2026 **Machine learning for Solid Earth Geosciences and Earthquake Physics (ML4SEG)** (Catania, Italy)

2026 **European Geosciences Union (EGU26)** [link] (*Wien, Austria*)

2025 **European Geosciences Union (EGU25)** [link] (*Wien, Austria*)

2024 **Deformation Mechanisms, Rheology and Tectonics (DRT24)** (*Barcelona, Spain*)

2024 **European Geosciences Union (EGU24)** [link] (*Wien, Austria*)

2024 **Tectonic Studies Group (TSG24)** (*St. Andrews, UK*)

2023 **Congresso congiunto SIMP, SGI, SOGEI, AIV** (*Potenza, Italy*)

2023 **European Geosciences Union (EGU23)** [link] (*Wien, Austria*)

Projects

- HighSeas.jl — SEAS simulations with High Performance Computing in mind
- Squiggles.jl — Julia package to accelerate 1D cross correlations of seismic signals using any GPU
- Fracability — Python toolbox to analyse fracture networks for digitalized rock outcrops
- Fractalizer.jl — Fractalize any shape
- SirGridsAlot — Lighting fast tilable regular grids in python

Skills

Languages	Italian (Native), English (Fluent)
Programming languages	Python, Julia, CUDA, MATLAB, Bash, CSS/HTML
Libraries	numpy, pandas/geopandas, shapely, VTK/pyvista, Makie.jl, CUDA.jl, DiffEq.jl
Software	Arc/QGIS, Agisoft Metashape, PZero/SKUA/Petrel/Move, Blender, FreeCAD/KiCAD
Technologies	Git, LaTeX, Markdown, SLURM, Docker, ssh